Data Science Project: Titanic Survival Prediction

Prepared by: Shreejit Mondal

Objective

The objective of this project is to apply the complete data science pipeline to the historical Titanic dataset. By walking through data analysis, visualization, and feature engineering, the ultimate goal was to construct a machine learning model capable of predicting passenger survival based on specific demographic and ticket information.

Steps Followed

1. **Data Collection:** Imported the Titanic dataset utilizing the Pandas and Seaborn libraries.
2. **Data Cleaning:** Handled missing values by imputing the median for the Age column to prevent skewing, and the mode for the Embarked column. Unnecessary columns with excessive missing data were dropped.
3. **Data Visualization:** Utilized Matplotlib and Seaborn to generate descriptive charts, including a survival bar plot, an age distribution histogram, and a correlation heatmap to understand relationships between variables.
4. **Feature Engineering:** Mapped categorical text data (Sex) to binary numeric formats. Engineered a new feature, FamilySize, by combining the SibSp (siblings/spouses) and Parch (parents/children) columns, plus 1 for the passenger themselves.
5. **Model Building:** Selected the requested features and split the dataset into an 80/20 training and testing ratio. A Logistic Regression classifier was then trained on the data.
6. **Evaluation:** Tested the model against the unseen test set, calculating the final accuracy score and generating a confusion matrix to evaluate false positives and false negatives.

Features Used

- **Age:** The age of the passenger in years.
- **Sex:** The gender of the passenger (mapped to 0 for male, 1 for female).
- **Pclass:** The ticket class (1st, 2nd, or 3rd).
- **Fare:** The amount paid for the ticket.
- **FamilySize:** An engineered feature representing the total number of family members aboard.

Machine Learning Model Logistic Regression was selected for this classification task. It is a highly effective baseline model for binary classification problems (Survived vs. Did Not Survive) and scales well with the numerical features prepared during the preprocessing phase.

Results & Conclusion The Logistic Regression model achieved an accuracy of approximately **81.01%** on the test data, falling comfortably within the target range of 75–85%.

The confusion matrix indicates that the model successfully identified the majority of true positives and true negatives. This project successfully demonstrates how foundational data science techniques—specifically data cleaning, targeted feature engineering, and logistic regression—can be utilized to extract meaningful patterns and make reliable predictions from historical datasets.